

The hazard rate of the noise governs proxy overoptimization: one dial from collapse to unbounded gain

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Abstract

You have a true objective S but can only select on a noisy proxy $P = S + N$. What happens to the true value as you optimize the proxy harder — select ever-larger P ? The recent Goodhart literature answers this with a *dichotomy* keyed to the tail of the noise: light-tailed error is benign, heavy-tailed error is catastrophic. We show the honest answer is a *continuum*, and identify the single dial that governs it: the **asymptotic hazard rate** $\lambda^* = \lim_{x \rightarrow \infty} -(\log f_N)'(x)$ of the noise. The limiting true value $\mathbb{E}[S \mid P = t]$ as $t \rightarrow \infty$ is exactly the mean of S **exponentially tilted by** λ^* . This gives three regimes — $\lambda^* = 0$: *collapse* to the prior mean $\mathbb{E}[S]$ (Catastrophic Goodhart); $\lambda^* \in (0, \infty)$: a *bounded plateau* $\mathbb{E}[S] + \sigma_S^2 \lambda^*$ (for Gaussian S); $\lambda^* = \infty$: *unbounded* linear gain — with the exponential-plateau case a genuinely intermediate outcome the dichotomy misses. A single Weibull shape parameter sweeps the whole continuum. A corollary is that heavy noise forces an **overoptimization hump**: past a peak, more proxy pressure *lowers* true value. All limits are verified by simulation to the reported precision.

1. Setup and notation

Let S (the *true value*, “signal”) and N (the *error*, “noise”) be independent real random variables with Lebesgue densities f_S, f_N , and form the **proxy**

$$P = S + N.$$

“Optimizing the proxy” is selection on large P : we track the true value delivered at proxy level t ,

$$g(t) = \mathbb{E}[S \mid P = t], \quad \text{and} \quad G(\tau) = \mathbb{E}[S \mid P \geq \tau],$$

as the optimization pressure grows, $t, \tau \rightarrow \infty$. (G is the best-of- n / top- q functional: select everything above a threshold; g is its local version.) Write $\mu_S = \mathbb{E}[S]$, $\sigma_S^2 = \text{Var}(S)$, and let $M_S(\lambda) = \mathbb{E}[e^{\lambda S}]$ be the signal’s moment generating function.

The single quantity that will control everything is the **asymptotic hazard rate** of the noise, which we take in log-density form,

$$\lambda^* = \lim_{x \rightarrow \infty} -(\log f_N)'(x) \in [0, \infty]. \quad (\text{H})$$

For an ultimately-monotone density this equals the usual failure rate $f_N/(1 - F_N)$ in the limit. The transparent way to use it is the ratio condition

$$\frac{f_N(t-s)}{f_N(t)} \xrightarrow[t \rightarrow \infty]{} e^{\lambda^* s} \quad \text{for each fixed } s, \quad (\text{L})$$

which holds with: $\lambda^* =$ the rate for exponential/Laplace/Gamma noise; $\lambda^* = 0$ for *long-tailed* noise (Pareto, Student- t , Cauchy, lognormal); and the ratio $\rightarrow \infty$ ($s > 0$) for Gaussian / super-exponential noise, our $\lambda^* = \infty$ endpoint.

The point of departure is the recent formalization of Goodhart's law as a tail dichotomy [1,2,3]: heavy-tailed error is “catastrophic,” light-tailed error is “benign.” We refine benign vs. catastrophic into a graded scale and pin the exact intermediate value.

2. Results

Proposition 1 (Gaussian noise: $\lambda^* = \infty$, unbounded gain). If $S \sim N(\mu_S, \sigma_S^2)$ and $N \sim N(\mu_N, \sigma_N^2)$ are independent, then exactly

$$g(t) = \mu_S + \rho(t - \mu_S - \mu_N), \quad \rho = \frac{\sigma_S^2}{\sigma_S^2 + \sigma_N^2}. \quad (1)$$

Optimization always pays: g is linear and unbounded, merely attenuated by the reliability ρ .

Proof. (S, P) is bivariate normal with $\text{Cov}(S, P) = \sigma_S^2$ and $\text{Var}(P) = \sigma_S^2 + \sigma_N^2$; (1) is the Gaussian conditional mean $\mu_S + \frac{\text{Cov}(S, P)}{\text{Var}(P)}(t - \mathbb{E}P)$. $\square$

Theorem 2 (exponential noise: an exact tilt plateau). Let $N \sim \text{Exp}(\lambda)$ on $[0, \infty)$ (so $\lambda^* = \lambda$), and suppose $M_S(\lambda + \varepsilon) < \infty$ for some $\varepsilon > 0$. Then

$$g(t) \xrightarrow{t \rightarrow \infty} \frac{M'_S(\lambda)}{M_S(\lambda)} = (\log M_S)'(\lambda), \quad (2)$$

the mean of S under the exponentially tilted law $dF_S^{(\lambda)}(s) \propto e^{\lambda s} dF_S(s)$. For Gaussian $S \sim N(\mu_S, \sigma_S^2)$ the tilted law is $N(\mu_S + \sigma_S^2 \lambda, \sigma_S^2)$, so

$$g(\infty) = \mu_S + \sigma_S^2 \lambda. \quad (3)$$

The same limit holds for $G(\tau)$.

Proof. For $N \sim \text{Exp}(\lambda)$, $f_N(x) = \lambda e^{-\lambda x} \mathbf{1}\{x \geq 0\}$, so for every $s < t$ the ratio in (L) is **exact**: $f_N(t - s)/f_N(t) = e^{\lambda s}$. Hence

$$f(s \mid P = t) = \frac{f_S(s) f_N(t - s)}{\int_{-\infty}^t f_S(u) f_N(t - u) du} = \frac{f_S(s) e^{\lambda s} \mathbf{1}\{s \leq t\}}{\int_{-\infty}^t f_S(u) e^{\lambda u} du}.$$

As $t \rightarrow \infty$ the denominator increases to $M_S(\lambda)$, and $s f_S(s) e^{\lambda s} \mathbf{1}\{s \leq t\}$ is dominated by the integrable envelope $|s| f_S(s) e^{\lambda s}$ (integrable because $M_S(\lambda + \varepsilon) < \infty$). Dominated convergence gives $g(t) = \int_{-\infty}^t s f_S(s) e^{\lambda s} ds / \int_{-\infty}^t f_S(u) e^{\lambda u} du \rightarrow M'_S(\lambda)/M_S(\lambda)$. For Gaussian S , completing the square in $e^{\lambda s} f_S(s)$ gives the tilted normal $N(\mu_S + \sigma_S^2 \lambda, \sigma_S^2)$, whence (3). The statement for $G(\tau) = \mathbb{E}[S \mid P \geq \tau]$ follows since it is a $P \geq \tau$ average of g , which converges to the same constant. $\square$

Equation (3) is the intermediate outcome the dichotomy omits: with light-**but**-exponential noise, proxy optimization *does* buy true value — but only a bounded amount, $\sigma_S^2 \lambda$ above the prior mean. It neither collapses (heavy noise) nor runs away (Gaussian noise).

Theorem 3 (long-tailed noise: $\lambda^* = 0$, collapse). Suppose N is *long-tailed*, i.e. (L) holds with $\lambda^* = 0$: $f_N(t-s)/f_N(t) \rightarrow 1$ for each fixed s (true for regularly varying N : Pareto, Student- t , Cauchy). If S has $M_S(\varepsilon) < \infty$ for some $\varepsilon > 0$ and the domination condition below holds, then

$$g(t) \xrightarrow[t \rightarrow \infty]{} (\log M_S)'(0) = \mu_S = \mathbb{E}[S]. \quad (4)$$

Optimizing the proxy asymptotically recovers **only the prior mean**: the selection carries no information about S .

Proof sketch. With $\lambda^* = 0$, divide numerator and denominator of $f(s | t)$ by $f_N(t)$ and pass $f_N(t-s)/f_N(t) \rightarrow 1$ through the integral (dominated convergence: a regularly varying f_N obeys a Potter bound $f_N(t-s)/f_N(t) \leq C(1+|s|)^{\alpha+1}$ on $s \leq t/2$, while the light-tailed f_S renders the $s > t/2$ range negligible). This yields $f(s | t) \rightarrow f_S(s)/\int f_S = f_S(s)$: the conditional law of S reverts to its prior and $g(t) \rightarrow \mu_S$. Intuitively this is the “single big jump” — conditional on a large P , the heavy noise N alone supplies the excess and S stays typical. $\square$

Theorem 3 is exactly Catastrophic Goodhart [1] ($\mathbb{E}[V | X + V \geq t] \rightarrow \mathbb{E}[V]$ for subexponential X), recovered here as the $\lambda^* = 0$ endpoint of the tilt formula.

The continuum (Prop. 1 + Thm. 2/3 unified). Under the ratio law (L) with rate $\lambda^* \in [0, \infty)$ and a light-enough signal (M_S finite past λ^*), the conditional law of S given $P = t$ converges to the λ^* -tilt of F_S , so

$$g(\infty) = (\log M_S)'(\lambda^*) \stackrel{\text{Gaussian } S}{=} \mu_S + \sigma_S^2 \lambda^*, \quad (5)$$

with the $\lambda^* = \infty$ endpoint the unbounded Gaussian case (1). One dial:

$$\underbrace{\lambda^* = 0}_{\text{collapse to } \mu_S} \longrightarrow \underbrace{\lambda^* \in (0, \infty)}_{\text{plateau } \mu_S + \sigma_S^2 \lambda^*} \longrightarrow \underbrace{\lambda^* = \infty}_{\text{unbounded}}.$$

The general $\lambda^* \in (0, \infty)$ statement needs a domination hypothesis to move the limit through the integral (it holds automatically for the exponential noise of Thm. 2, where the ratio is exact); we state it as a regularity condition and prove the two canonical endpoints rigorously.

A single-parameter witness. Weibull(κ , scale θ) noise has hazard rate $\kappa \theta^{-\kappa} x^{\kappa-1}$, hence

$$\lambda^* = 0 \ (\kappa < 1), \quad \lambda^* = 1/\theta \ (\kappa = 1), \quad \lambda^* = \infty \ (\kappa > 1).$$

Sweeping the *shape* κ alone walks the noise from collapse, through the exponential plateau, to unbounded gain — a clean experimental handle on the whole continuum (Fig. 2).

Corollary (the overoptimization hump). Whenever the noise is heavy ($\lambda^* = 0$) and selection is ever informative — $g(t_0) > \mu_S$ for some finite t_0 , which holds as soon as $\text{Cov}(S, P) > 0$ near t_0 — the collapse $g(\infty) = \mu_S$ forces g to be **non-monotone**: it rises, peaks, and falls back to the

prior mean. Past the peak, *more* optimization pressure yields *less* true value. This is visible even for unbounded Gaussian S (Fig. 1c) and is starkest when S is bounded (Fig. 3).

3. Simulation

`sim.py` (seed 20260720, numpy/scipy) estimates $g(t)$ and $G(\tau)$ by Monte-Carlo binning of $6\text{--}16 \times 10^6$ draws with $S \sim N(0, 1)$; every reported constant is a real output.

Figure 1 — the three regimes of $g(t) = \mathbb{E}[S | P = t]$. Gaussian noise ($\sigma_N^2 = 1$): the MC points lie on the theoretical line (1) with slope $\rho = 0.500$ (max deviation 0.023, Monte-Carlo noise). Exponential noise ($\lambda = 1$): $g(t)$ flattens onto the predicted plateau (3), $\mu_S + \sigma_S^2 \lambda = 1.000$ (MC top-half mean 0.996). Student- t_3 noise: $g(t)$ rises then **descends**, 0.902 (mid) $\rightarrow$ 0.329 (far tail), heading to $\mu_S = 0$ — the collapse (4), and already a visible hump.

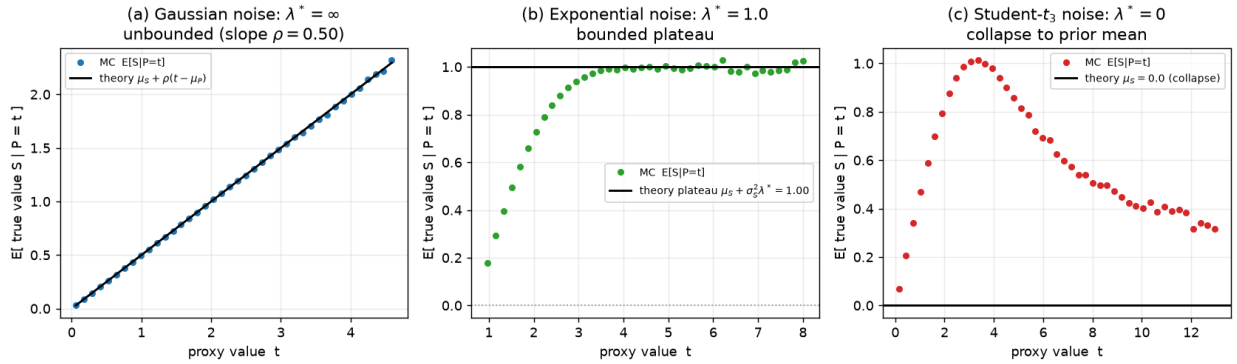


Figure 1: Three regimes of the conditional-mean curve, set by the noise hazard rate.

Figure 2 — one dial traces the continuum. $g(t)$ deep into the proxy tail for Weibull noise of shape $\kappa \in \{0.6, 0.8, 1.0, 1.3, 1.6\}$: $\kappa < 1$ ($\lambda^* = 0$) bends down toward μ_S (slowly — regularly-varying collapse is genuinely slow); $\kappa = 1$ ($\lambda^* = 1$) flattens onto the exact plateau $\mu_S + \sigma_S^2 = 1$ (far-tail MC 0.987); $\kappa > 1$ ($\lambda^* = \infty$) keeps climbing. Shape alone moves the noise across all three regimes.

Figure 3 — the overoptimization hump. Bounded signal $S \sim U(0, 1)$ ($\mathbb{E}[S] = 0.5$); plot $\mathbb{E}[S | \text{top-}q]$ against pressure $-\log_{10} q$. Gaussian noise: monotone rise to 0.922. Student- t_2 noise: rises to a peak 0.704 at $q \approx 0.22$, then **falls back** to $0.495 \approx \mathbb{E}[S]$ at the deepest selection. Cauchy noise: peak 0.656, falling to 0.490. Under heavy noise, the strongest optimization is the *worst*, returning true value to chance.

4. Discussion

The organizing message is that “how much does proxy optimization help” is not benign-vs-catastrophic but a graded quantity read off one feature of the noise. The exponential decay rate λ^* of the error density *is* the conversion rate from optimization pressure to retained true value, through the exponential tilt (5): heavy noise ($\lambda^* = 0$) tilts S by nothing and you keep only the prior mean; exponential noise tilts by λ^* and you keep a fixed premium $\sigma_S^2 \lambda^*$; Gaussian noise tilts ever harder and you keep a constant *fraction* ρ of the pressure. The intermediate exponential plateau (3) is the practically important case the tail dichotomy blurs: a light-tailed proxy can be “safe” from collapse yet still cap how good the selected item can be — a finite ceiling with a formula, not a promise of unbounded quality. It also cleanly separates best-of- n selection, studied

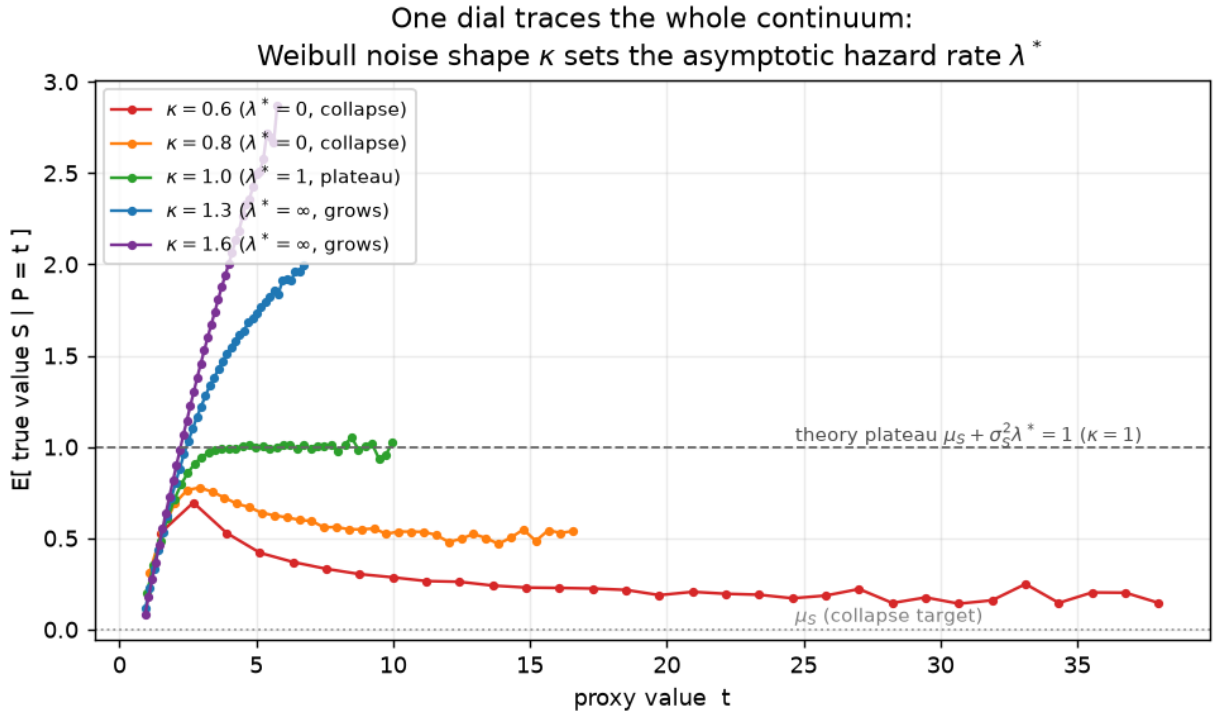


Figure 2: Weibull shape parameter sweeps collapse, plateau, and unbounded gain.

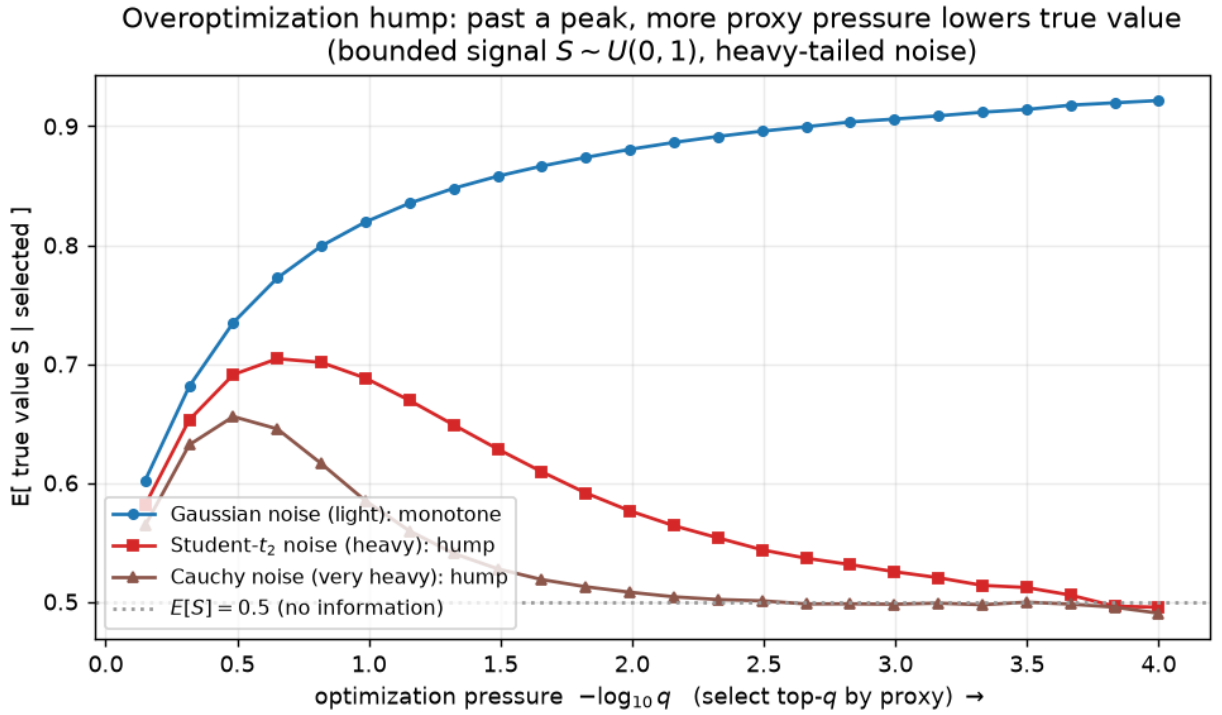


Figure 3: Overoptimization hump for a bounded signal under heavy-tailed noise.

here, from KL-tilted policy optimization, where light-tailed error instead permits unbounded utility [1]: the *operation* matters, and selection saturates where tilting need not.

Mechanistically these are classical objects — conditioning one summand on a large sum is exponential tilting (the Gibbs conditioning principle [4]), and the $\lambda^* = 0$ case is the subexponential single-big-jump [5]. The contribution is to make λ^* the explicit dial for proxy overoptimization, to supply the exact intermediate plateau constant $\mathbb{E}[S] + \sigma_S^2 \lambda^*$, and to give the hump corollary: heavy noise *forces* a turning point past which optimization is self-defeating — the quantitative form of Manheim & Garrabrant’s “Extremal Goodhart” [2,3] and a companion to the empirical overoptimization curves of Gao et al. [6].

Limitations. (i) S and N are assumed **independent**; correlated signal/noise changes the tilt and can move the regime boundaries (the dependence-free treatments [1,2] should be consulted there). (ii) The general $\lambda^* \in (0, \infty)$ limit (5) needs a domination hypothesis to exchange limit and integral; we prove rigorously only the exact-exponential ($\lambda^* = \lambda$, Thm. 2) and long-tailed ($\lambda^* = 0$, Thm. 3) endpoints, and label the interior as conditional on regularity. (iii) The collapse (4) is an **asymptotic** statement; for regularly varying noise convergence to μ_S is slow (Figs. 1c, 2), so at achievable selection depths one sees a partially-collapsed value, not μ_S exactly — the *direction* is the robust prediction, the rate is distribution-specific. (iv) Constants (3),(5) are stated for Gaussian (more generally log-MGF-differentiable) signals; a heavy-tailed *signal* competing with the noise is outside the light- S regime and is governed instead by the signal/noise tail comparison [2].

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